File permissions in Linux

Project description

Scenario,

In this scenario, examine and manage the permissions on the files in the `/home/researcher2/projects` directory for the `researcher2` user.
The `researcher2` user is part of the `research_team` group.

check the permissions for all files in the directory, including any hidden files, to make sure that permissions align with the authorization that should be given. When it doesn't, you must change the permissions.

Here's how you'll do this task: **First,** check the user and group permissions for all files in the projects directory. **Next,** check whether any files have incorrect permissions and change the permissions as needed. **Finally,** check the permissions of the `/home/researcher2/projects/drafts` directory and modify these permissions to remove any unauthorized access.

Check file and directory details

file structure of the `/home/researcher2/projects` directory and the permissions of the files and subdirectory it contains.

In the `/home/researcher2/projects` directory, there are five files with the following names and permissions:

- `project_k.txt`
  - User = read, write,
  - Group = read, write
  - Other = read, write
- `project_m.txt`
  - User = read, write
  - Group = read
  - Other = none
- `project_r.txt`
  - User= read, write
  - Group = read, write
  - Other = read
- `project_t.txt`
  - User = read, write
  - Group = read, write
  - Other = read
- `.project_x.txt`
  - User = read, write
  - Group = write
  - Other = none

There is also one subdirectory inside the `projects` directory named `drafts`. The permissions on `drafts` are:

- User = read, write, execute
- Group = execute
- Other = none

```
researcher2@5fb54954501f:~$ ls
projects
researcher2@5fb54954501f:~$ cd projects
researcher2@5fb54954501f:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Jul  5 09:27 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Jul  5 09:27 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Jul  5 09:27 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jul  5 09:27 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jul  5 09:27 project_t.txt
```

To check files and directory details the `ls -l` command has been used.

Describe the permissions string

“drwxrwxrwx” is the 10 characters string which indicates the permission for user, group and others. Where the first character indicates the file type “d” for directory could also be the “-” for regular files
r - read
w - write
x - execute

From 2nd to 4th character indicates the permission for users. when one of these characters is a hyphen “-” instead, it indicates that this permission is not granted to the user

The 5th- 7th character indicates the permission for the group. when one of these characters is a hyphen “-” instead, it indicates that this permission is not granted to the group

The 8th-10th character indicates the permission for the others. when one of these characters is a hyphen “-” instead, it indicates that this permission is not granted to the others

A directory with full permission for all owners type would be “drwxrwxrwx”

Change file permissions

```
researcher2@5fb54954501f:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Jul  5 09:27 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Jul  5 09:27 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Jul  5 09:27 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jul  5 09:27 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jul  5 09:27 project_t.txt
researcher2@5fb54954501f:~/projects$ chmod o-w project_k.txt
researcher2@5fb54954501f:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Jul  5 09:27 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Jul  5 09:27 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Jul  5 09:27 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jul  5 09:27 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jul  5 09:27 project_t.txt
```

`chmod` is used to change the file permission, it stands for change mode.
here,
`Chmod o-w project_k.txt` is used to change the permission for project_k.txt, where write permission has been removed from other users to mitigate the security risk.

Change file permissions on a hidden file

```
researcher2@5fb54954501f:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jul  5 09:27 .
drwxr-xr-x 3 researcher2 research_team 4096 Jul  5 10:04 ..
-rw--w---- 1 researcher2 research_team   46 Jul  5 09:27 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jul  5 09:27 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Jul  5 09:27 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Jul  5 09:27 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jul  5 09:27 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jul  5 09:27 project_t.txt
researcher2@5fb54954501f:~/projects$ chmod u-w,g-w .project_x.txt
researcher2@5fb54954501f:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jul  5 09:27 .
drwxr-xr-x 3 researcher2 research_team 4096 Jul  5 10:04 ..
-r-------- 1 researcher2 research_team   46 Jul  5 09:27 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jul  5 09:27 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Jul  5 09:27 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Jul  5 09:27 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jul  5 09:27 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jul  5 09:27 project_t.txt
```

To view the hidden files `ls -la` command is used,
And to change the hidden files permission `chmod` command has been used to remove write permission for users and group to apply least privilege principle in order to protect the organisation

Change directory permissions

```
researcher2@5fb54954501f:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jul  5 09:27 .
drwxr-xr-x 3 researcher2 research_team 4096 Jul  5 10:04 ..
-r--r----- 1 researcher2 research_team   46 Jul  5 09:27 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jul  5 09:27 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Jul  5 09:27 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Jul  5 09:27 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jul  5 09:27 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jul  5 09:27 project_t.txt
researcher2@5fb54954501f:~/projects$ chmod g-x drafts
researcher2@5fb54954501f:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jul  5 09:27 .
drwxr-xr-x 3 researcher2 research_team 4096 Jul  5 10:04 ..
-r--r----- 1 researcher2 research_team   46 Jul  5 09:27 .project_x.txt
drwx------ 2 researcher2 research_team 4096 Jul  5 09:27 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Jul  5 09:27 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Jul  5 09:27 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jul  5 09:27 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jul  5 09:27 project_t.txt
researcher2@5fb54954501f:~/projects$
```

Here, the drafts directory permission has been changed by removing executing permission from group using `chmod g-x drafts`

Summary

Authorization is the concept of granting access to specific resources in a system. It's important because without authorization any user could access and modify all files belonging to other users or system files. This would certainly be a security risk.

In Linux, file and directory permissions are used to specify who has access to specific files and directories

As a security analyst, setting appropriate access permissions is critical to protecting sensitive information and maintaining the overall security of a system.